

Unit: 4 Trees, Planar graphs and ①

colouring of a graph.

Trees, Planar graphs, Euler's formula, dual of a planar graph, independence number and clique number, chromatic number, statement of Four-color theorem.

Applications: Scheduling and resource allocation in networks.

Trees:

Acyclic graph: A graph is said to be acyclic if it contains no cycles.

A tree is a connected acyclic graph.

An acyclic graph is called forest.

Examples:



Properties of trees:

* Any two vertices are connected by a unique path

* A tree with 'n' vertices has 'n-1' edges.

Theorem: 1 If in a graph G , there is exactly one path between every pair of vertices, then G is a tree. (2)

Proof: Existence of a path between every pair of vertices shows that G is connected.

A circuit in a graph implies, there is at least one pair of vertices 'a', 'b' such that there are two distinct paths between 'a' & 'b'. But there is only one path between every pair of vertices.

$\Rightarrow G$ has no circuit

$\Rightarrow G$ is a tree.

Theorem: 2 A tree with 'n' vertices has $(n-1)$ edges.

Proof: The Theorem will be proved by induction on 'n' = no. of vertices. For $n=1, 2, 3$, the Theorem is true.


No edge


1-edge


2-edges

Assume that the Theorem is true (3)
for $(n-1)$ vertices.

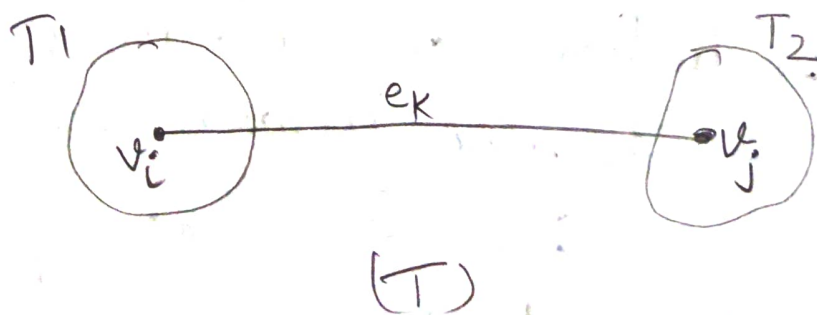
Consider a tree T with n vertices.

In T , let e_k be an edge with
end vertices v_i and v_j . Hence there is
only one path e_k between v_i and v_j .
 $\therefore$ deletion of e_k from T will disconnect
the graph.

Also $T - e_k$ consists of exactly two
components T_1 and T_2 . Since there is no circuit
in T , each of these components is a tree.

Both the trees T_1 & T_2 have less than
 n vertices and therefore by induction
each tree T_1 & T_2 contains $n-1$ edges.

Thus $T - e_k$ consists of $n-2$ edges.
Together with e_k , T has exactly $(n-1)$ edges.



Theorem: 3 If any tree with 4
2 (or) more vertices, there are at least
2 pendent vertices.

Proof: Assume a tree T with n
vertices. Hence $|E(T)| = n - 1$.

By Handshaking Theorem, total degree

of a tree is $2(n - 1) = 2n - 2$.

Prove that, T contains at least 2
Pendent Vertices.

If not, i.e., no 2 vertices of degree 1,
each vertex is of degree ≥ 2 . Since
we have n vertices, degree of
all the vertices in T is $\geq 2n$,
which is a contradiction.

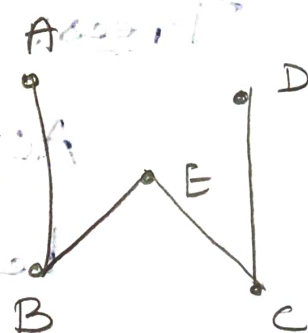
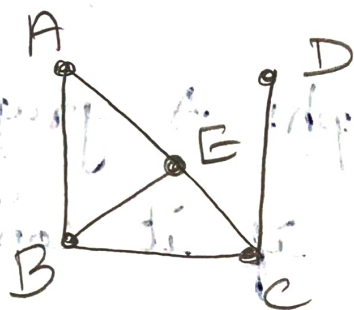
Hence T contains at least 2
Vertices of degree 1.

Spanning tree:-

(5)

A spanning tree for a connected graph G is a tree containing all the vertices of G and a minimum number of edges of G .

Example:



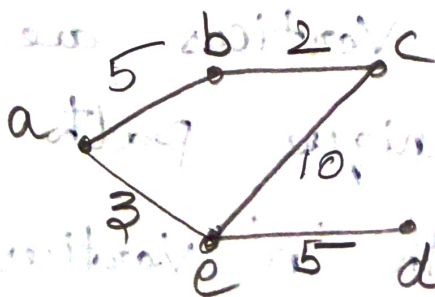
(6)

Spanning tree
of G .

Weighted graph:

A graph is said to be a weighted graph if each edge is assigned weights (a number).

Ex:



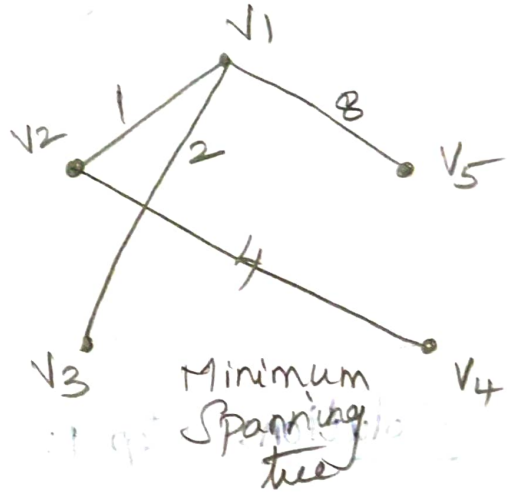
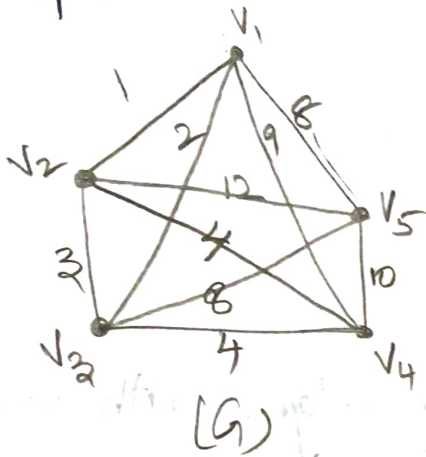
Weighted graph

Minimum Spanning Tree: (MST)

(6)

A spanning tree of minimum weight in a connected weighted graph G is called a minimum spanning tree of G .

Example:



Kruskal's algorithm: to find MST.

Input: A weighted connected graph G

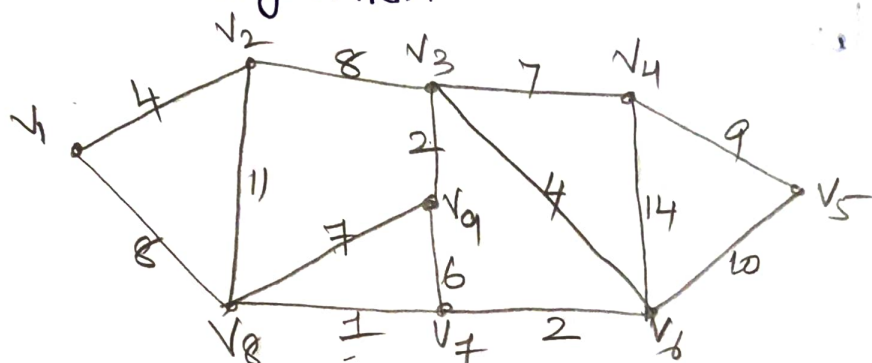
Output: A minimum spanning tree T of G .

Step 1: Arrange all the edges in increasing order of their weights.

Step 2: Pick the smallest edge. Add it to the vertices of the spanning tree, if it does not form a cycle. If cycle is formed discard it.

Step 3: Repeat Step 2, until there are $n-1$ edges in the spanning tree.

1) Find the MST for the connected weighted graph given below using Kruskal algorithm



Solution: Step 1: List of edges with weights in increasing order.

$V_8, V_7 - 1$

$V_1, V_8 - 8$

$V_3, V_9 - 2$

$V_2, V_3 - 8$

$V_6, V_7 - 2$

$V_4, V_5 - 9$

$V_3, V_6 - 4$

$V_5, V_6 - 10$

$V_1, V_2 - 4$

$V_2, V_8 - 11$

$V_9, V_7 - 6$

$V_4, V_6 - 14$

$V_3, V_4 - 7$

$V_8, V_9 - 7$

Step 2:

V_2

V_3

V_4

V_9

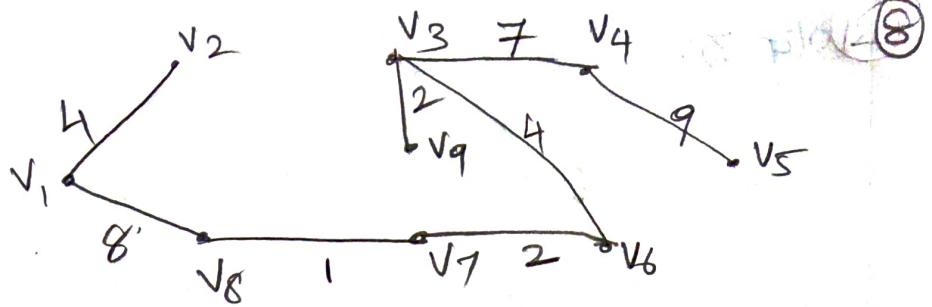
V_5

V_8

V_7

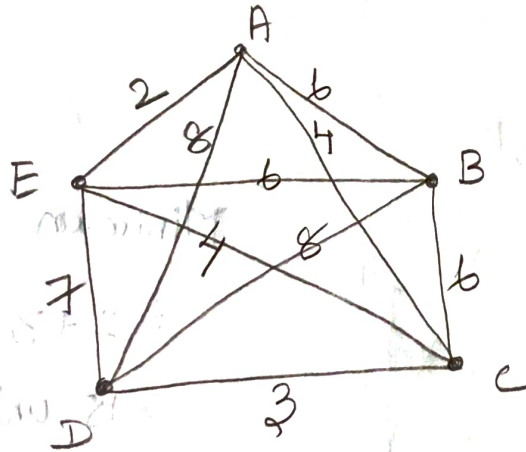
V_6

Step 2:



Minimum spanning tree with
weight. $= 1 + 2 + 2 + 4 + 4 + 7 + 8 + 9$
 $= 37$ units.

2) Find the MST for the given weighted graph using Kruskal algorithm.



Step 1:

Sol List of edges with weights in increasing order.

AE - 2

CD - 3

AC - 4

EC - 4

BC - 6

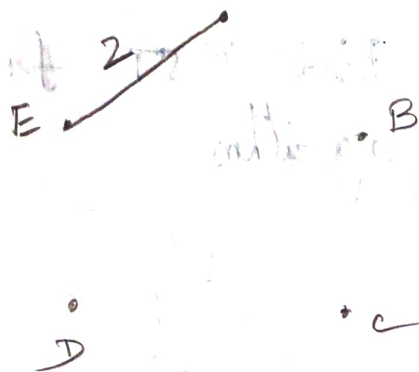
BE - 6

DE - 7

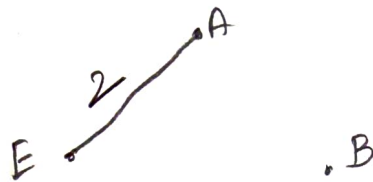
AD - 8

DB - 8

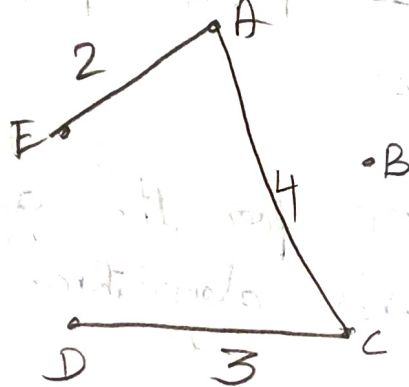
Step 2:



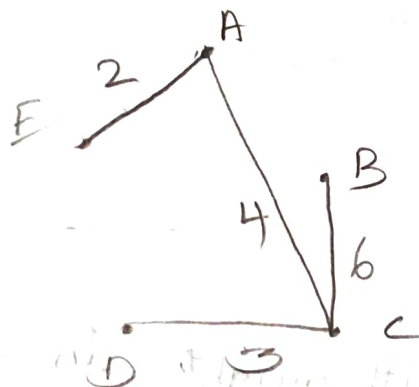
Step 3:



Step 4:



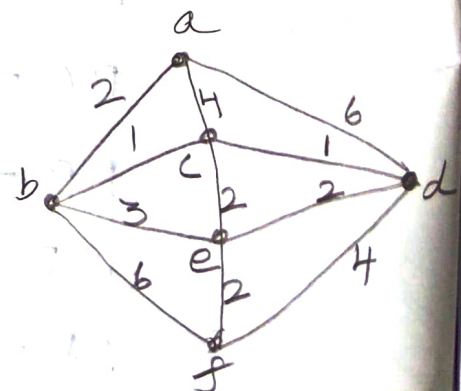
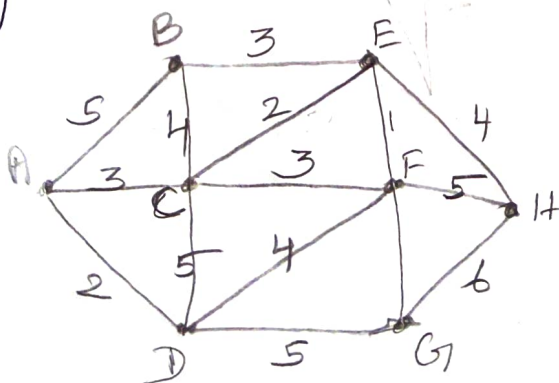
Step 5:



Minimum weight
 $= 2 + 3 + 4 + 6$
 $= 15 \text{ units}$

Practise Problem:

- 1) Find MST for the followings using Kruskal algorithm



Prim's algorithm:

(10)

Input: A weighted connected graph G

Output: A minimum spanning tree.

Let $|V(G)| = n$.

Step 1: choose any starting vertex

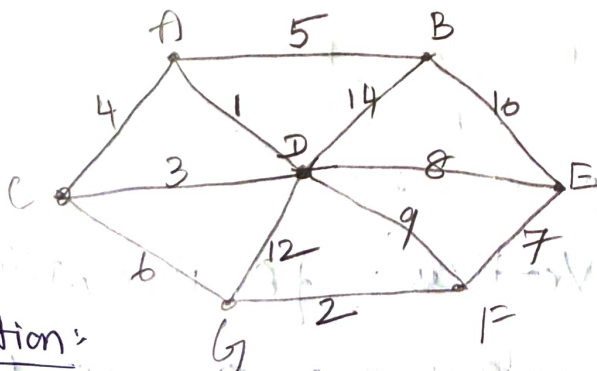
Step 2: Find all the edges that are incident to it.

Step 3: Select an edge with minimum weight

Step 4: Add the edge and the new vertex to the MST

Step 5: Repeat step 2-4 until all vertices are included.

1) Find the minimum spanning tree for the given weighted graph using Prim's algorithm.



Solution:

Step 1: Choose A as the starting vertex

MST Vertices: A

Possible edges: AB - 5, AC - 4
AD - 1,

Choose the minimum: AD - 1



Step 2: MST Vertices: A, D

Possible edges:

AB - 5

AC - 4

DB - 14

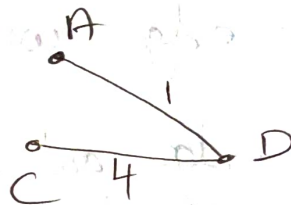
DC - 3

DE - 7

DF - 9

DG - 12

Choose the minimum: AC - 4.



Step 3: MST Vertices: {A, D, C}

Possible edges:

AB - 5

DF - 9

DB - 14

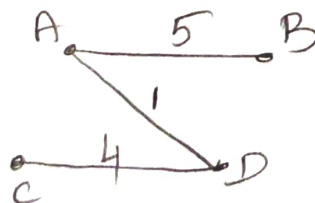
DG - 12

~~DE - 7~~

CG - 6

DE - 7

Choose the minimum: AB - 5



Step 4: MST Vertices: {A, B, C, D}

Possible edges:

DE - 8

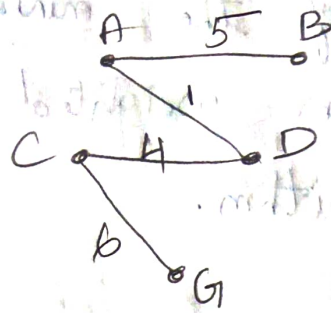
CG - 6

DF - 9

BE - 10

DG - 12

Choose the minimum: CG - 6



Step 5: MST Vertices $\{A, B, C, D, G\}$

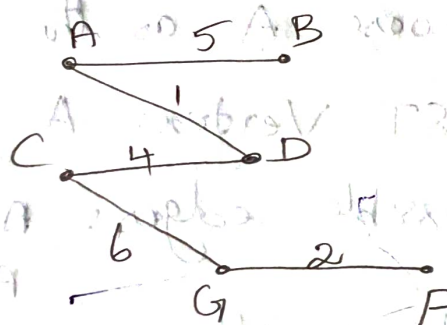
Possible edges: ~~DE-8~~

~~DF-9~~

~~BE-10~~

GF-2

Choose the minimum: GF-2



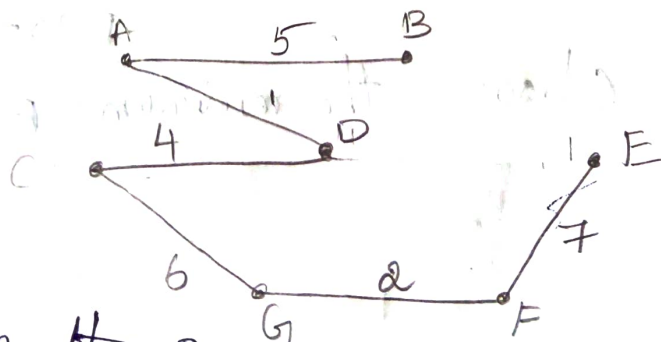
Step 6: MST Vertices $\{A, B, C, D, G, F\}$

Possible edges: ~~DE-8~~

~~BE-10~~

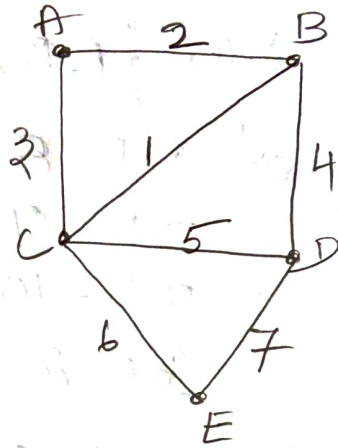
FE-7

Choose the minimum: FE-7



Weight of the Minimum spanning tree $= 1 + 4 + 5 + 6 + 2 + 7 = 25$ units

2) Find the weight of the minimum spanning tree for the given weighted graph using Prim's algorithm. (B)



Solution.

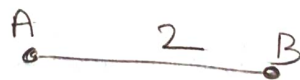
Step 1: Choose A as the starting vertex.

MST Vertex: A

Possible edges: AB - 2

AC - 3

Choose the minimum: AB - 2



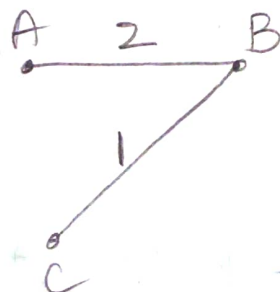
Step 2: MST Vertices: {A, B}

Possible edges: AC - 3

BC - 1

BD - 4

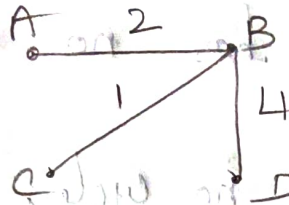
Choose the minimum: BC - 1



Step 3: MST Vertices: $\{A, B, C\}$ (14)

Possible edges: $BD - 4$
 $CD - 5$
 $CE - 6$

Choose the minimum: $BD - 4$

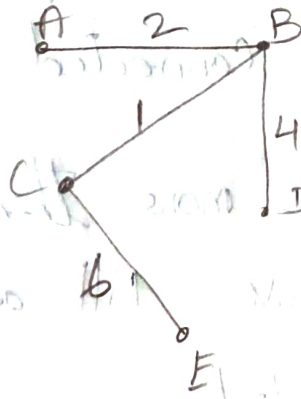


Step 4: MST Vertices: $\{A, B, C, D\}$

Possible edges: $CE - 6$

$DE - 7$

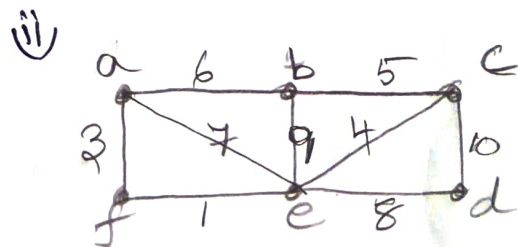
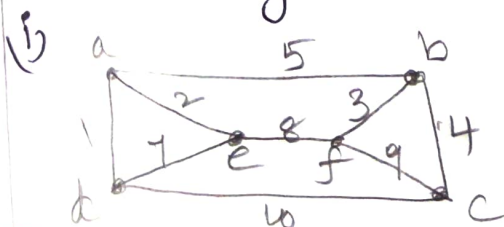
Choose the minimum: $CE - 6$



All the Vertices are included in MST

Total weight = $2 + 1 + 4 + 6 = 13$

Practice questions: Find the MST using Prim's algorithm.



Theorem: 1 Every connected graph has 15
atleast one spanning tree

Proof: let G be a connected graph

Case (i) If G has no cycle.

Since G has no cycle, G itself is its own spanning tree.

Case (ii) If G has a circuit.

Delete an edge from the circuit.
Now, G becomes connected and circuitless

If G has more than one circuit, repeat the above till an edge from last circuit is deleted.

$\therefore G$ is connected without any cycle that contains all the vertices of G .

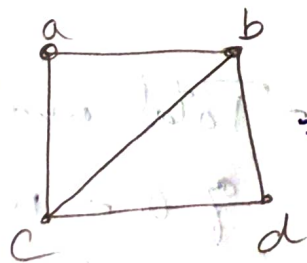
$\Rightarrow G$ is a spanning tree.

Branch: An edge in a spanning tree is ⁽¹⁶⁾ called a branch.

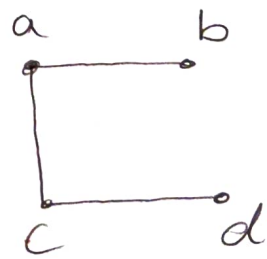
Chord: Let G be a connected graph and T be the spanning tree of G . An edge which is in G , but not in T is called a chord.

Note: with respect to any of its spanning tree, a connected graph with ' n ' vertices and ' e ' edges has $(n-1)$ branches and $e-n+1$ chords.

Ex:



One of its spanning tree $T =$



No of Branches = $ab, ac, cd = n-1 = 4-1 = 3$

No of Chords = $e-n+1 = 5-4+1 = 2 (bc, bd)$

Theorem: 2 Let G be a connected graph with n vertices and e edges and T be its spanning tree. Then T has $(n-1)$ branches and $(e-n+1)$ chords.

Solution: Let G be a connected graph with n vertices and e edges. Let T be the spanning tree of G .

Since T is a spanning tree with n vertices, T has $(n-1)$ edges. Hence number of branches $= n-1$.

$$\begin{aligned}\text{Number of chords} &= \text{Total number of edges of } G - \text{number of branches} \\ &= e - (n-1) \\ &= e - n + 1.\end{aligned}$$

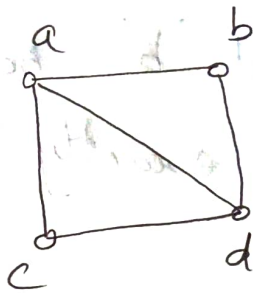
Fundamental circuits:

Let G be a connected graph and T be its spanning tree. Any circuit which is formed by adding a chord

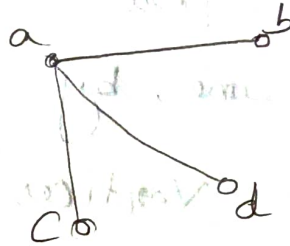
to its spanning tree is called a 18
fundamental circuit.

Eg:

$G =$



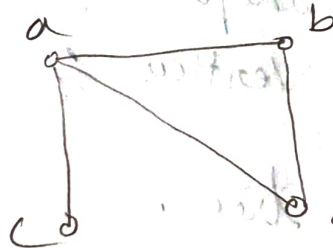
Spanning tree $T_1 =$



Here, chords are cd, db . Adding the

Chord bd to T_1

$T_2 =$



The circuit $a-b-d-a$ is called a
fundamental circuit.

Theorem: 3 Any connected graph G
is a tree if and only if adding an edge
in G , creates exactly one circuit.

Proof: Necessary Part:

Let G be a tree.

$\Rightarrow$ There exist a unique path between
any pair of vertices. say a' and b' .

If we add an edge between a and b in G , then the unique path between a and b becomes a circuit.
 $\Rightarrow G$ has exactly one circuit.

Sufficient Part

Assume, by adding an edge between any two vertices a and b of a connected graph G , G has exactly one cycle.

$\Rightarrow G$ has unique path between every pair of vertices of G .

$\Rightarrow G$ is a tree.

Rank of a graph:

(20)

The number of branches in any spanning tree T of a connected graph G is called the rank, denoted by r .

$$r = \text{No of branches}$$

Note: For any spanning tree (T) of a connected graph, the number of edges is $n-1$. $\therefore r = n-1$

Nullity of a graph

The number of chords in G is called nullity of G , denoted by μ .

$$\mu = e - (n-1)$$

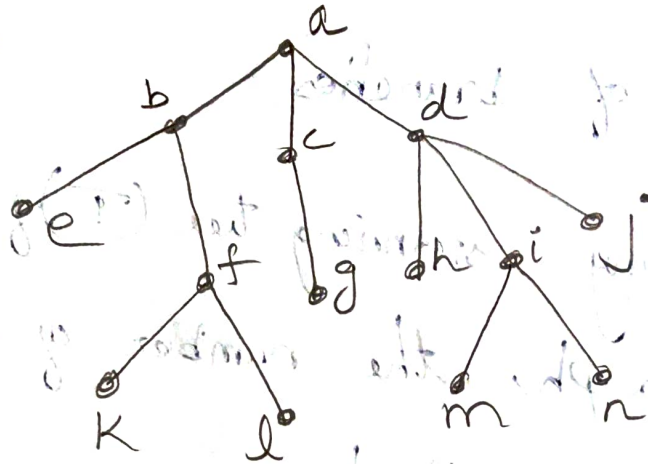
where $e = \text{no of edges in } G$
 $n = \text{No of vertices in } G$

Note: For any connected graph G ,
 $\text{rank} + \text{nullity} = |E(G)|$

Rooted tree.

A rooted tree is a tree in which a particular vertex is distinguished from the others and it is called the root.

Ex:



Rooted tree with the root vertex as 'a'.

Internal Vertices:

The set of vertices other than degree 1 (pendent) vertices are called internal vertices.

Levels in a rooted tree.

The zero level — Root vertex.

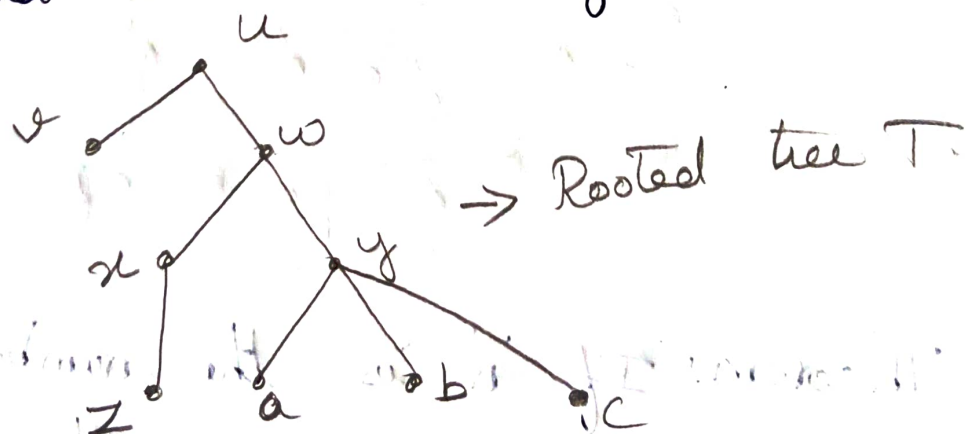
1st level — Vertices adjacent to the root vertex.

2nd level - Vertices adjacent to 1st level
vertices. ~~except~~ ~~in~~
and so on. (22)

Childrens of a Vertex in rooted tree.

Consider a vertex v in a rooted tree. The Childrens of the Vertex ' v ' are the vertices which are adjacent to v in its next level.

- 1) Consider the rooted tree T
- (i) what is the root of T ?
 - (ii) Find the leaves and internal vertices?
 - (iii) What is the level of w and z ?
 - (iv) Find the children of w and y ?



Solution:

- (i) root = u
- (ii) leaves = u, z, a, b, c .
internal vertices = w, x, y .
- (iii) Level of $w = 1$, level of $z = 3$.

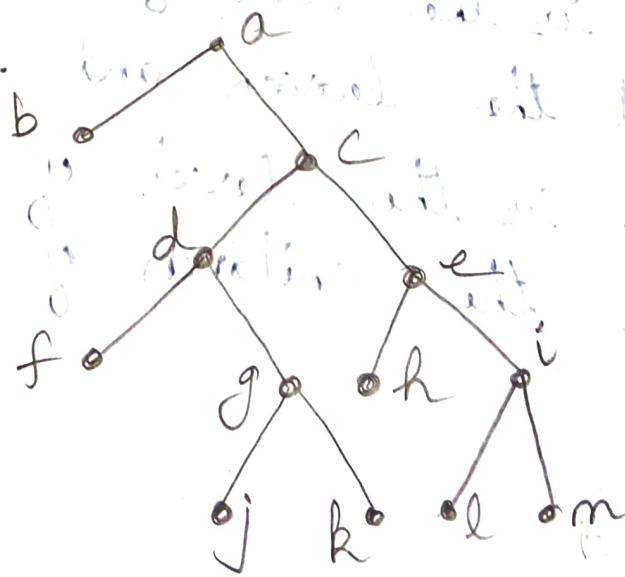
(iv) childrens of $w = x, y$ and
childrens of $y = a, b, c$

Binary tree

A binary tree is a tree in which exactly one vertex of degree two and other vertices of degree 1 (or) 3.

Note: In a binary tree, the vertex of degree 2 is called as root.

Example:



Theorem: If n is the number of vertices of a binary tree, then

- (1) n is odd
- (2) The number of pendent vertices is $n+1$

(3) The number of Vertices of degree 2 is $\frac{n-3}{2}$. (24)

(4) The number of internal Vertices is $\frac{n-1}{2}$.

Proof: Let T be a binary tree with n Vertices. Then by def T has only one Vertex of degree two and other Vertices of degree 1 or 3.

$\Rightarrow$ there are $(n-1)$ odd degree Vertices

We know that, the number of odd degree Vertices is even.

So, $n-1$ is even

$\Rightarrow n$ is odd

② Let P be the number of Pendent Vertices and m be the number of Vertices of degree 3.

$$m = n - P - 1$$

Hence, the sum of the degree of all the Vertices of the tree

$$= (1 \times 2) + (P \times 1) + (n - P - 1) \times 3$$

$$= 3n - 2p - 1$$

By Handshaking Theorem, sum of the degree of all the vertices is equal to 2 times the number of edges.

$$\Rightarrow \underline{\text{No}} \text{ of edges} = \frac{1}{2} (3n - 2p - 1)$$

$$\Rightarrow n - 1 = \frac{1}{2} (3n - 2p - 1)$$

$$\Rightarrow 2n - 2 = 3n - 2p - 1$$

$$\Rightarrow p = \frac{n+1}{2} \quad \text{--- (2)}$$

③ No of vertices of degree 2 is

$$m = n - p - 1$$

$$m = n - \left(\frac{n+1}{2}\right) - 1 \quad (\text{By (2)})$$

$$m = \frac{2n - n - 1 - 2}{2}$$

$$m = \frac{n-3}{2}$$

④ The number of internal vertices is

$$= n - p$$

$$= n - \left(\frac{n+1}{2}\right)$$

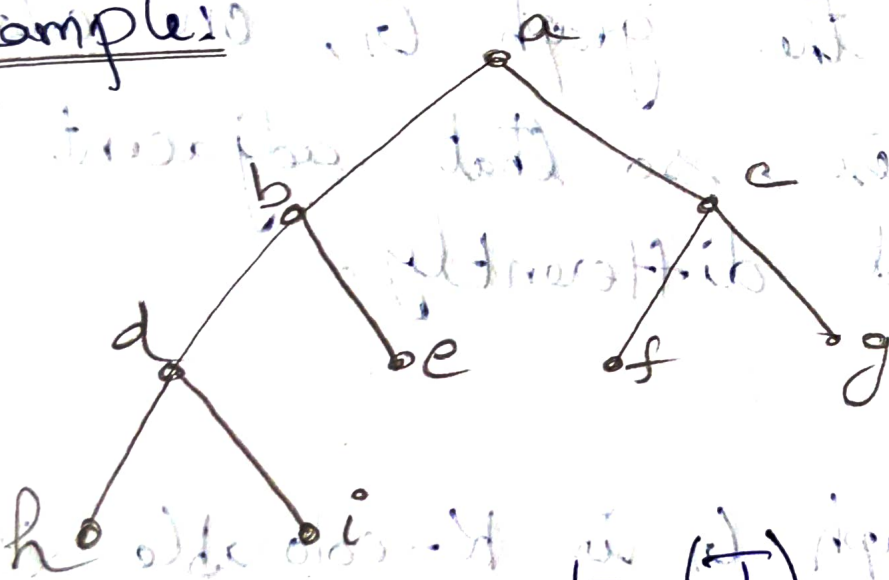
$$= \frac{n-1}{2}$$

Height of a binary tree:

(26)

The ^{maximum} number of edges from the root vertex to a pendent vertex.

Example



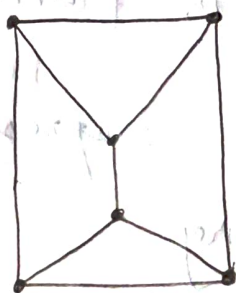
Binary tree (T)

Height of T = 3. (a-b-d-h)

Planar graph

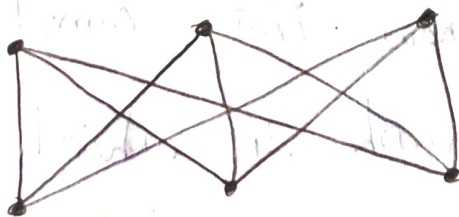
(27)

A graph G is said to be planar if there exists some geometric representation of G which can be drawn on a plane such that no two edges intersect.

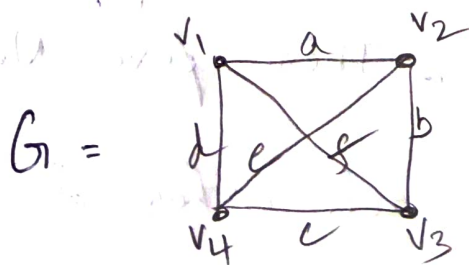


Non-planar graph:

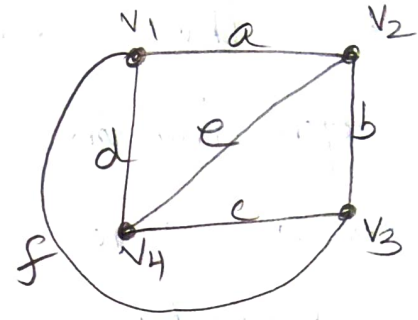
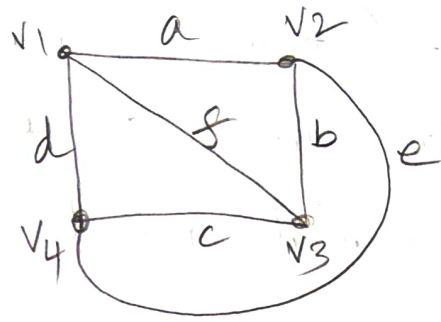
A graph that cannot be drawn on a plane without a cross over between its edges is called Non-planar.



Example: Consider the graph G .



The geometric representation such that no two edges intersect are. (28)



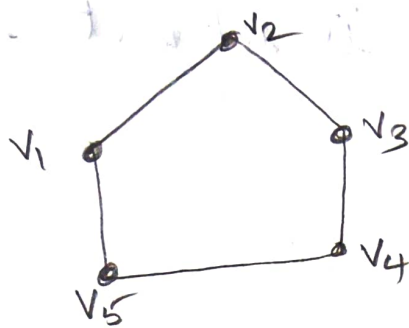
$\therefore G$ is Planar

Note: (1) Geometric representation is called embedding
(2) Planar graph means it should have Planar embedding.

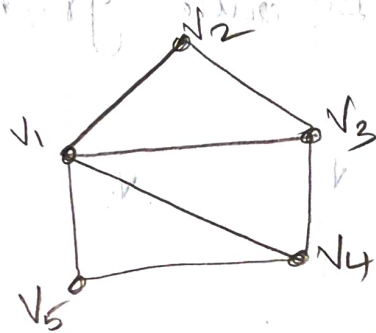
Problems: 1) Prove that a complete graph on 5-Vertices is non-Planar.

Proof: We know that complete graph ~~is~~ is a simple graph in which there is an edge between every pair of vertices.

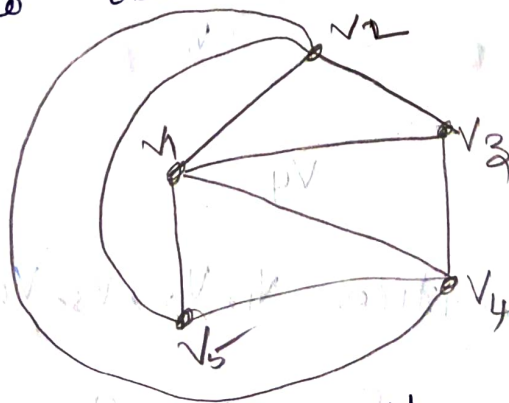
Let the 5 vertices of K_5 be v_1, v_2, v_3, v_4 and v_5 . These five vertices are connected to each other to form a cycle.



Now, V_1 should be adjacent with V_3 and V_4 . Both the edges can be drawn within the cycle or outside the cycle. Let us draw it internally.



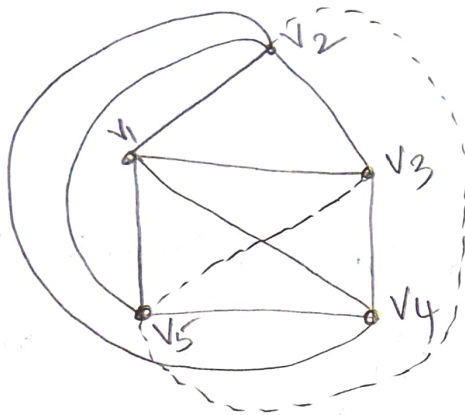
Now, the vertex V_2 should be adjacent with V_5 and V_4 . They cannot be drawn inside the cycle, since it intersects the edges connecting V_1V_3 and $V1V4$. So we have to draw externally.



Next, the vertex V_3 should be adjacent to the vertex V_5 . If we draw inside it intersects the edge $V1V4$ and if

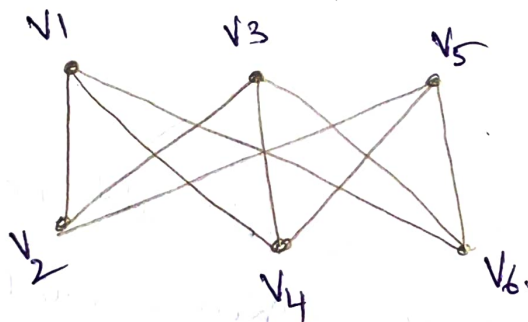
we draw outside, it intersect the edge (3)

$V_2 V_4$.



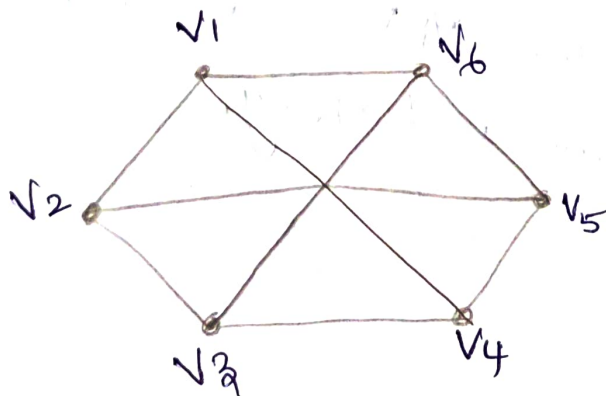
Thus K_5 is non-Planar graph.

2) The complete bipartite graph $K_{3,3}$ is non-Planar.



Solution:

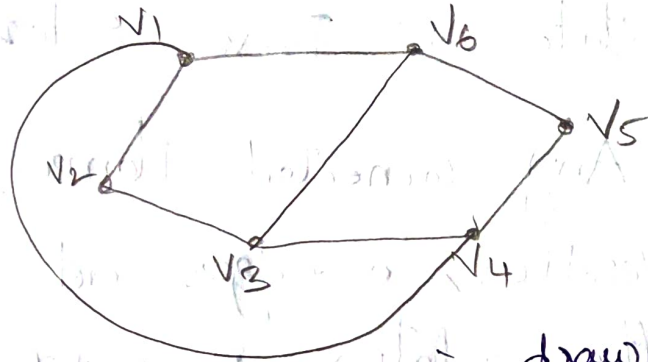
The (isomorphic) representation of $K_{3,3}$ is



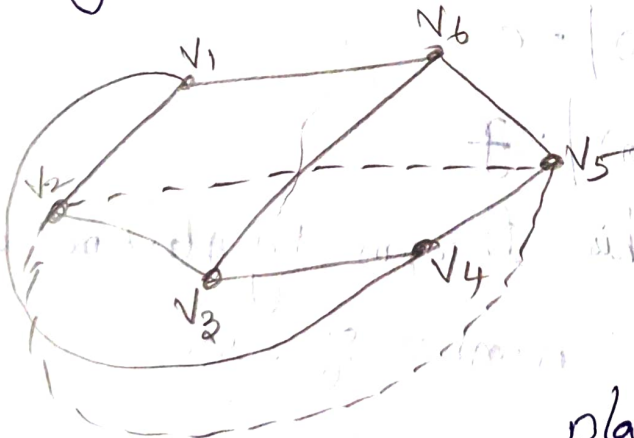
The six vertices V_1, V_2, V_3, V_4, V_5 and V_6 form a hexagon. There are 9 edges. Out of these 9 edges, 6 edges are used for the hexagon sides. The remaining

3-edges should be drawn inside or outside the hexagon. (31)

The edge V_3V_6 can be drawn inside without cross over and the edge V_1V_4 can be drawn outside without cross over.



If the edge N_2V_5 is drawn inside or outside the hexagon, it crosses over already drawn edge.



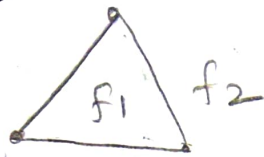
Thus $K_{3,3}$ is non-planar.

Definition. A face is an enclosed region formed by the vertices and edges of a Planar graph when it is drawn on a flat surface without any edges crossing. This also includes the single, unbounded region

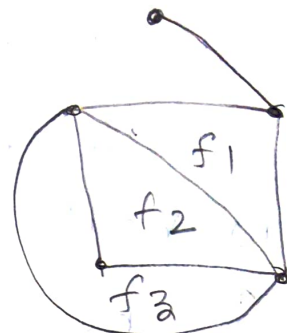
that surrounds the entire graph (the $\textcircled{3}$ outer face).

Ex:

K_3 :



f_4



Theorem: State and Prove Euler's formula.

Statement: Any connected planar graph with ' n ' vertices, ' e ' edges and ' f ' faces satisfies the relation $n - e + f = 2$.

Proof: Let G be a graph with

$$|V(G)| = n$$

$$|E(G)| = e$$

$$|F(G)| = f$$

We prove this theorem by mathematical induction on number of edges.

Step 1: Let $e = 0$ Then $n = 1$ and $f = 1$

$$\text{Hence } n - e + f = 1 - 0 + 1 = 2.$$

Step 2: Assume Euler's formula is true for every connected planar graph having fewer than e edges.

Step 2: Prove Euler's formula is true (33)
for a connected Planar graph G
with e edges.

There are two cases:

Case 1: G is a tree. Then $|V(G)| = n$,
 $|E(G)| = n-1$, $|F(G)| = 1$.

$$\therefore n - e + f = n - (n-1) + 1 = 2.$$

Case 2: G contains at least one cycle.

Remove an edge from the cycle.

Removing this edge

(i) does not disconnect the graph

(ii) decreases the number of edges by 1

(iii) merges two adjacent faces into one

$$\therefore |V(G)| = n, |E(G)| = e-1, |F(G)| = f-1$$

Since the resulting graph has fewer edges

by step 2: $n - (e-1) + (f-1) = 2$

$$n - e + f = 2.$$

Thus, Euler's formula holds.

Corollary: In any simple, connected graph with f -regions, n -vertices and e -edges ($e \geq 2$), the following inequalities hold
(i) $e \geq \frac{3}{2}f$ (ii) $e \leq 3n - 6$

Proof: In a connected planar graph, every region is bounded by at least 3 edges.

$\therefore$ Sum of degrees of all faces $\geq 3f$.

Each edge is counted twice.

$$\therefore 2e \geq 3f.$$

$$\Rightarrow e \geq \frac{3}{2}f. \text{ --- (1)}$$

Using Euler's formula

$$e \geq \frac{3}{2}(e - n + 2)$$

$$\Rightarrow e \geq \frac{3}{2}e - \frac{3}{2}n + 3$$

$$\Rightarrow \frac{3}{2}e - e \leq \frac{3}{2}n - 3$$

$$\Rightarrow \frac{1}{2}e \leq \frac{3}{2}n - 3$$

$$\Rightarrow e \leq 3n - 6.$$

Problem: ① Prove that K_5 is non-planar. (35)

Sol: K_5 has $n=5$ vertices and $e=10$ edges. Suppose if K_5 is planar, then by Euler's formula

$$e \leq 3n - 6$$

$$\Rightarrow 10 \leq 3 \times 5 - 6$$

$$\Rightarrow 10 \leq 9 \quad (\text{which is not possible})$$

$\therefore K_5$ is non-planar

② Prove that $K_{3,3}$ is non-planar.

Sol: $K_{3,3}$ has 6 vertices and 9 edges.

Suppose $K_{3,3}$ is planar.

In $K_{3,3}$, no region can be bounded with fewer than 4 edges. Hence the number of edges bounding f faces is $\geq 4f$.

Since, edges of $K_{3,3}$ lie in the boundary of 2-region, $2e \geq 4f$

Using Euler's formula

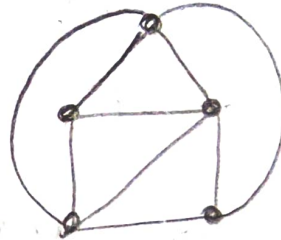
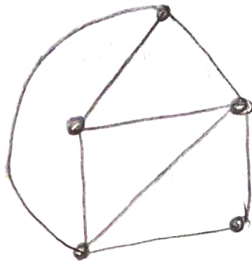
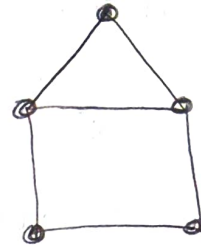
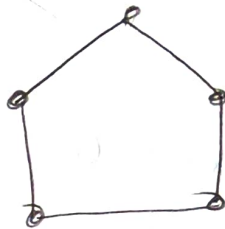
$$2e \geq 4(e - n + 2)$$

$$2 \times 9 \geq 4(9 - 6 + 2)$$

$$18 \geq 20 \quad (\text{which is not possible})$$

Hence $K_{3,3}$ is non-planar.

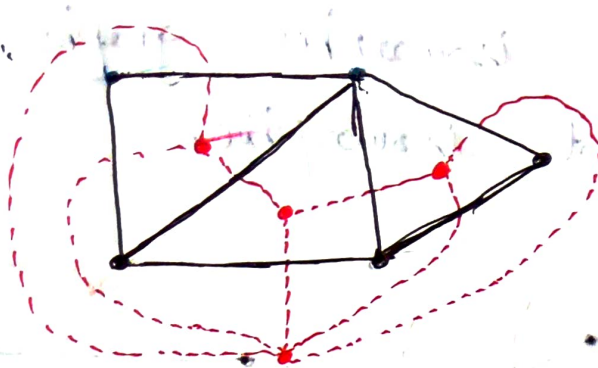
③ Draw all planar graphs with 5 vertices, which are not isomorphic to each other.



Dual graph(G^*)

The dual graph, of a planar graph G is a graph that has a vertex for each face of G , and has an edge for each pair of faces in G that are separated from each other by an edge, a self-loop when the same face appears on the both side of an edge. It is denoted by G^* .

Example:

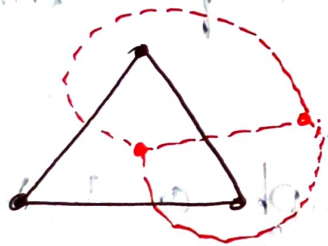


(37)

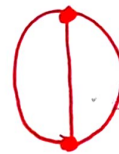
The red graph is the dual of the black graph and vice versa.

Problems

① Find the dual for K_3 .



G

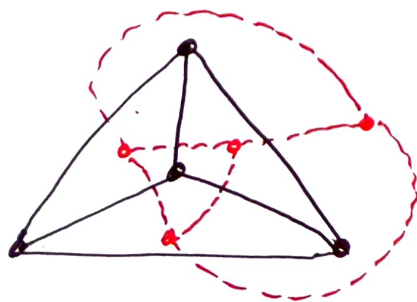


G^*

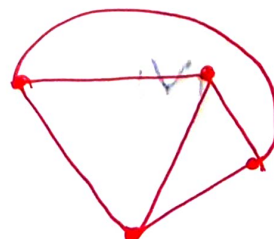
Self-dual graphs:

A plane graph is said to be self-dual if it is isomorphic to its dual graph.

Example:



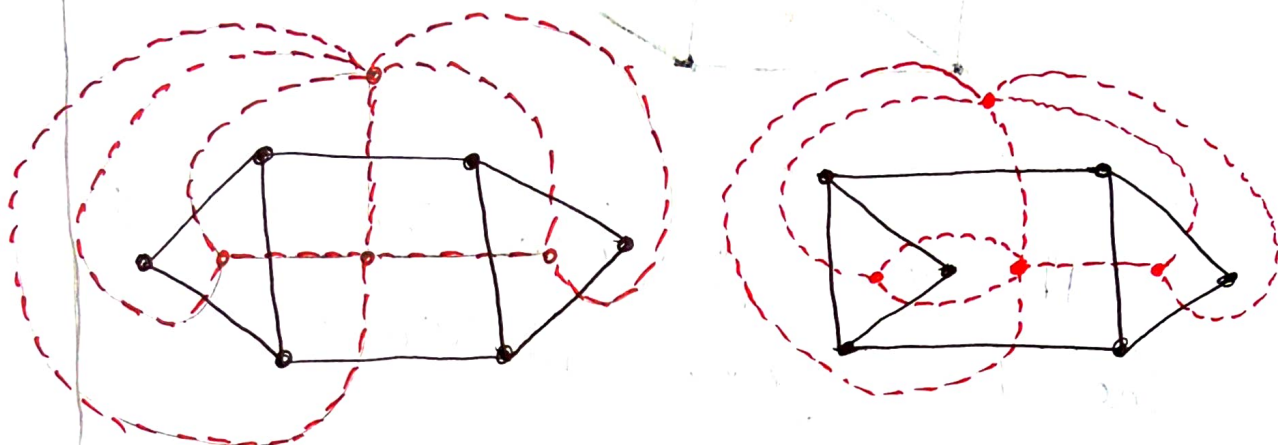
G



G^*

$$G \cong G^*$$

② Draw two isomorphic graphs whose dual are not isomorphic. 3*



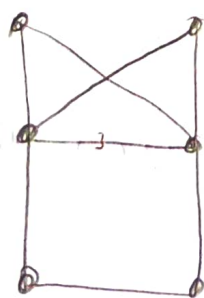
Practice Problems..

1) Draw the dual for the following graphs.

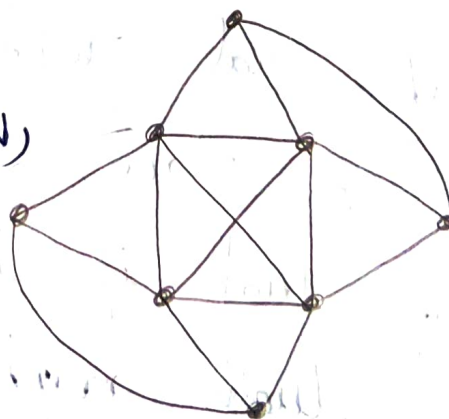
(i) Wheel graph on 5 vertices

(ii) $K_{3,2}$

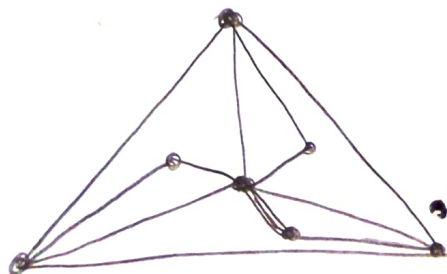
(iii)



(iv)



(v)



Independent set:

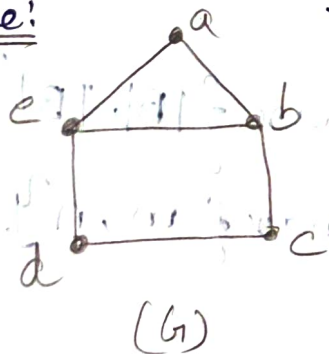
(39)

A set $I \subseteq V(G)$ is called an independent set if no two vertices in I are adjacent.

Independence number of G :

The parameter $\alpha(G) = \max\{|I| : I \text{ is an independence set of } G\}$ is called the independence number of G .

Example:



Independence set

$\{a\}, \{b\}, \{c\}, \{d\}, \{e\},$
 $\{a, d\}, \{a, c\}, \{e, c\}$
 $\{b, d\}, \{c, e\}.$

Independence number $\alpha(G) = 2$.

① Find $\alpha(G)$ for complete graph K_n .

Sol. W.K.T, in a complete graph each vertex is adjacent to all the other vertices. Hence $\alpha(K_n) = 1$.

2) Find $\alpha(K_{m,n})$.

Sol. Let $G(V, E) = K_{m,n}$ be a complete bipartite graph with $V = A \cup B$, where

$$|A|=m, |B|=n, A \cap B = \emptyset. \quad (40)$$

Also, every vertex in A is adjacent to every vertex in B , and no two vertices within A and B are adjacent.

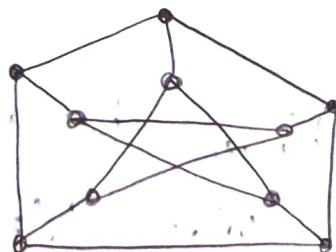
Since there are no edges connecting vertices within A , the set A itself is an independent set of size m .

Similarly, B is an independent set of size n .

$$\alpha(K_{m,n}) = \max\{|A|, |B|\} \\ = \max\{m, n\}.$$

Practice Problems:

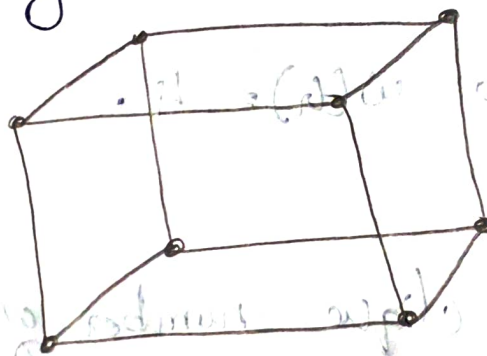
- 1) Find $\alpha(G)$ for cycle graph C_n (for both even and odd n).
- 2) Find $\alpha(G)$ for path graph P_n .
- 3) Find all possible independent set and independent number for the given graph



(Petersen graph)

4) Construct a simple graph G with $|V(G)| = 7$ such that G is bipartite with maximum degree $\Delta(G) = 4$ and G contains no isolated vertices. Find the independence number for the constructed graph.

5) Find $\alpha(G)$, where G is Hypercube Q_3 which is given below:



clique number:

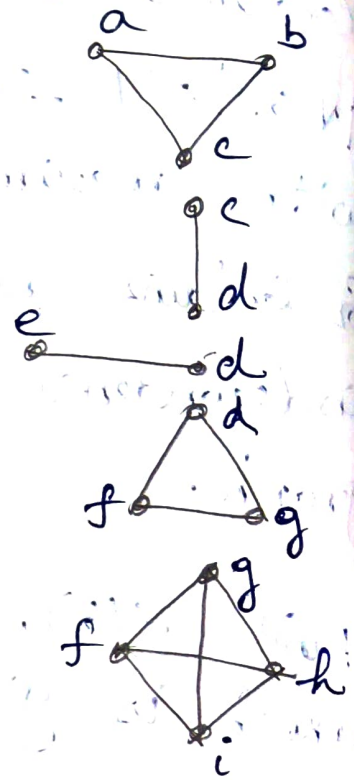
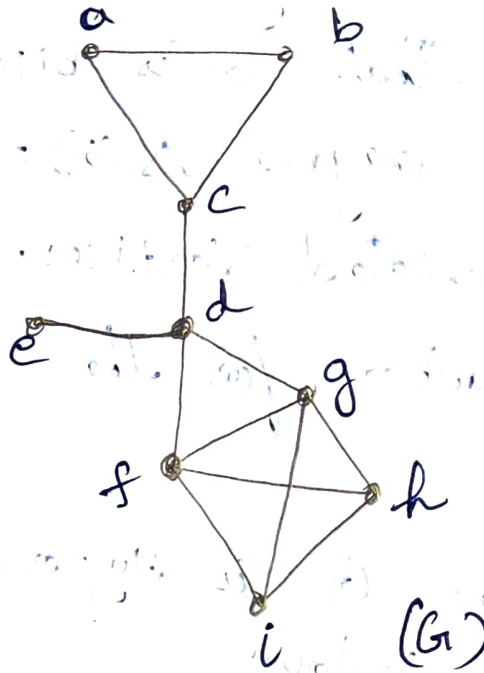
Clique: A subgraph $\mathcal{Q}$ of a graph G is called a clique if any two vertices in $\mathcal{Q}$ are adjacent.

clique number:

The clique number of G is the integer $\omega(G) = \max \{ |V(\mathcal{Q})| : \mathcal{Q} \text{ is a clique in } G \}$

Example:

clique:



Clique number $\omega(G) = 4$.

Problems:

1) Find the clique number of K_n .

Sol:

In a complete graph K_n , every single vertex is adjacent to every other vertices. The graph itself form the largest clique. $\therefore \omega(K_n) = n$.

2) Find $\omega(C_n)$.

Sol:

For C_3 , $\omega(C_3) = 3$, because it form a triangle.

For $n \geq 4$, $\omega(C_n) = 2$, since a cycle (43)

contains connected pairs of adjacent vertices (clique of size 2) but no 3-vertex loop (triangle).

2) Find the clique number of bipartite graph.

Sol: A bipartite graph divides vertices into two sets with edges only running between the sets. It is impossible to form a K_3 . So the largest clique size is 2. $\omega(G) = 2$, where G is a bipartite graph.

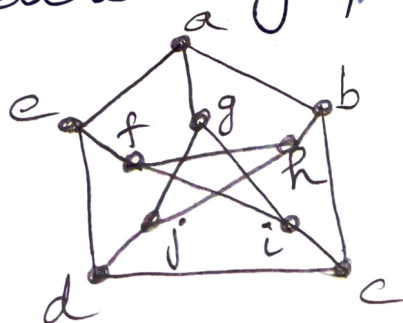
Practice Problems.

Find $\omega(G)$ for

(i) Trees (T)

(ii) wheel graphs (W_n)

(iii) Petersen graph



Coloring of a graph.

(44)

Proper Vertex coloring of a graph.

An assignment of colors to the vertices of the graph G , one color to each vertex, so that adjacent vertices are colored differently.

K-colorable:

A graph G is K -colorable if there exist a coloring of G from a set of K -colors.

Chromatic number:

The minimum positive integer K for which G is K -colorable is the chromatic number, denoted by $\chi(G)$.

K-chromatic graph

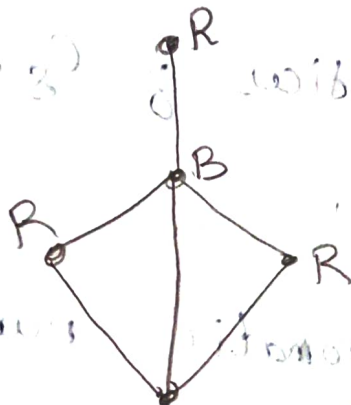
A graph G with chromatic number K is called K -chromatic graph.

Note: 1) If H is a subgraph of G ,
then $\chi(H) \leq \chi(G)$

2) A graph with only isolated vertices is 1-chromatic. (45)

3) A graph with at least one edge is at least 2-chromatic.

Example:

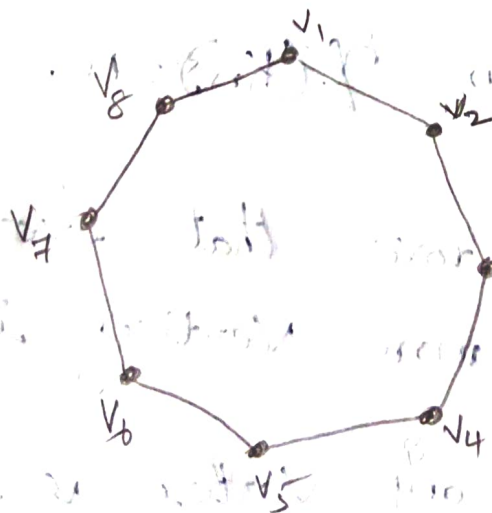


G. is 3-chromatic graph

Problems:

1) Find the chromatic number of cycles with 8 vertices (C_8)?

Sol:



$\Rightarrow C_8$

Start coloring from the vertex v_1 with the color R. Next vertex v_2 cannot receive the same color, hence color v_2 with B.

v_3 is adjacent with v_2 and v_4 . Hence v_3 is colored using B . Hence v_3 is colored using B . Likewise, $v_4 \neq B$, $v_5 = R$, $v_6 = B$, $v_7 = R$, $v_8 = B$. Hence we used two colors (R, B) to properly color the vertices of C_8 . Hence $\chi(C_8) = 2$.

2) Find the chromatic number of the complete graph K_n .

Sol: In K_n , each vertex is adjacent to every other vertices. Hence every vertex should be uniquely colored. Thus $\chi(K_n) = n$.

Theorem: 1 Prove that every tree with 2 (or) more vertices is 2-chromatic.

Proof: Select any vertex v in a tree T and let it be the root of T . Color the root by R . Color the vertices adjacent to v by B .

color 2. Now, color all the vertices which are adjacent ^{to} ~~with~~ color 2. Vertices with 1.

(47)

Proceed similarly till every vertex in the given tree T is colored.

Now, in T , we have two cases.

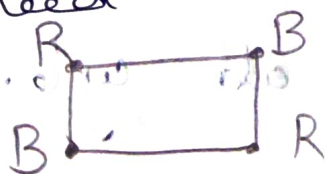
Case (i): All the vertices at odd distance from v have color - 2

Case (ii): The vertex v and all the vertices at even distance from v have color - 1.

Since, there is a unique path in T , no two adjacent vertices have same color. Thus T is properly colored with 2 colors. Hence, every tree is 2-chromatic.

Note: Converse of the above theorems need not be true. i.e., every 2-chromatic graph need not be a tree.

Ex



$\Rightarrow$ 2-chromatic but not a tree.

Theorem: 2 (König's theorem)

(48)

A graph with at least one edge is 2-chromatic iff it has no cycles of odd length.

Proof: Necessary part:

Assume G is 2-chromatic

$\Rightarrow G$ has no cycle of odd length.

Suppose not,

ie, G has a cycle of odd length.

$v_1, v_2, \dots, v_n$. Then $v_1, v_3, v_5, \dots, v_n$ has color 1, $v_2, v_4, v_6, \dots, v_{n-1}$ has color 2.

Since cycle is odd, n^{th} vertex and 1st vertex are adjacent ie, v_1 and v_n are adjacent receiving the same color. So, either v_1 or v_n should be colored with 2.

$\Rightarrow G$ is 3-chromatic

which is a contradiction to our assumption.

$\therefore G$ has no odd cycle.

Sufficient Part

(49)

Assume G contains cycle of even length.

Construct a spanning tree T of

G . Since, every tree is 2-chromatic,
 $\Rightarrow$ Spanning tree is 2-chromatic.

Now add the chords (edges in G but not in T) to T one by one.

Since G has no cycle of odd length the end vertices of every chord being replaced are differently colored in T . Thus G has proper coloring with 2-colors.

$\Rightarrow G$ is 2-chromatic.

Theorem: 3 Prove that the

chromatic number of a graph will

not exceed by more than one the

maximum degree of the vertices in

a graph.

(or)

Let $\Delta(G)$ be the maximum degree

of the graph G , then $\chi(G) \leq 1 + \Delta(G)$.

(50)

Proof: Suppose $\chi(G) > 1 + \Delta(G)$

$$\Rightarrow \chi(G) \geq \Delta(G) + 2$$

$$\Rightarrow \chi(G) - 1 \geq \Delta(G) + 1$$

We know that, there are at least χ vertices each of degree at least

$$\chi - 1.$$

$\therefore$ There are at least χ vertices each of degree $\geq \chi - 1$, which is a contradiction, that no vertex can be of degree $> \Delta$.

$\therefore \chi > \Delta(G) + 1$ is wrong.

$$\Rightarrow \chi \leq \Delta(G) + 1.$$

Theorem 4: A graph is 2-chromatic if and only if it is bipartite.

Proof: Necessary Part:

Assume G is 2-chromatic.

$\Rightarrow V(G)$ can be partitioned into ⁽⁵¹⁾ two disjoint independent sets.

Since, it is 2-chromatic, G does not contain an odd cycle.

Hence, G is bipartite.

Sufficient part:

Assume G is bipartite.

P.T G is 2-chromatic.

By the definition of bipartite graph, $V(G)$ is partitioned into two disjoint independent subsets. Also, G does not contain an odd cycle.

$\Rightarrow G$ is 2-chromatic.

Four Color Problem:

Every map (ie., a planar graph) can be properly colored with 4-colors
(or)

Every planar graph has 4-colorable.
